23 MOVEMENT AND SUPPORT IN MAMMALS OBJECTIVE QUESTIONS AND ANSWERS

Each question in this section is followed by four options A to D. Find the correct option for each question.

1. What happens when the biceps muscle contracts? The
   A. forearm straightens.
   B. triceps muscle contracts.
   C. scapula moves towards the sternum.
   D. forearm bends.

   [SSCE 1993 Qu 10]

2. All the vertebrae of mammals have the following **except**
   A. neural canal.
   B. neural spine.
   C. facets for articulation.
   D. vertebrarterial canals.

   [SSCE 1995 Qu 49]

3. The appendicular skeleton is made up of the
   A. skull and vertebral column.
   B. ribs and vertebral column.
   C. limbs and limb girdles.
   D. lumbar and thoracic vertebrae

   [SSCE 1995 Qu 57]

4. The substance that lubricates bones at the joint is called
   A. synovial fluid.
   B. lymph.
   C. sebum.
   D. endolymph.

   [SSCE 1996 Qu 45]

5. Which of the vertebrae has the most developed centrum?
   A. Caudal vertebrae
   B. Cervical vertebrae
   C. Lumbar vertebrae
   D. Thoracic vertebrae

   [SSCE 1997 Qu 18]

6. Which of the following explains why the lumbar vertebra of mammals has a well-developed centrum?
   A. To reduce the number of vertebrae
   B. For adequate protection of the spinal cord
   C. To bear the weight of the body
   D. For the attachment of transverse processes

   [SSCE 1998 Qu 53]

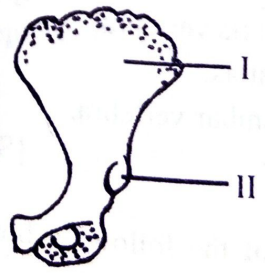
7. The precise location of the atlas is
   A. between the skull and the axis.
   B. the front of the rib cage.
   C. between the last lumbar vertebrae and the sacrum.
   D. within the pelvic girdle.

   [SSCE 1999 Qu 37]

8. The acetabulum articulates with the
   A. femur.
   B. humerus.
   C. rib.
   D. odontoid process

   [SSCE 1999 Qu 51]

Study the diagram below and use it to answer Questions **9** *to* **11**.



9. The bone shown in the diagram above is the
   A. coccyx.
   B. pelvis.
   C. scapula.
   D. sternum.

   [SSCE 2000 Qu 21]

10. The part labelled **I** is the
A. ilium.
B. ischium.
C. scapula.
D. sacrum.
[SSCE 2000 Qu 22]

11. The bone that articulates with the part **II** is the
A. femur.
B. fibula.
C. humerus.
D. tibia.
[SSCE 2000 Qu 23]

12. The significance of the odontoid process is that it
A. provides a point for pivoting.
B. strengthens the vertebra.
C. provides points for the attachment of muscles.
D. allows the spinal cord to join the brain.
[SSCE 2000 Qu 47]

13. Friction is reduced at movable joints by
A. ligaments and cartilage.
B. tendons and synovial fluid.
C. synovial fluid and cartilage.
D. ligaments and tendons.
[SSCE 2000 Qu 59]

14. A vertebra with a long spine pointing backwards is likely to be
A. a thoracic vertebra.
B. an axis vertebra.
C. an atlas.
D. a lumbar vertebra.
[SSCE 2001 Qu 18]

15. Which of the following bones forms part of the axial skeleton?
A. Atlas
B. Fibula
C. Humerus
D. Scapula
[SSCE 2001 Qu 29]

16. The structure that provides a point for pivoting the atlas is the
A. anterior hypapophysis.
B. odontoid process.
C. posterior hypapophysis.
D. transverse process.
[SSCE 2002 Qu 6]

17. Friction-free movement at the knee joint is enhanced by the presence of
A. cartilage.
B. ligament.
C. synovial fluid.
D. tendon.
[SSCE 2002 Qu 7]

18. Friction is reduced at joints because
A. bones are held firmly by ligaments.
B. synovial fluid is present between the bones.
C. the bones are separated by gaps.
D. cartilage covers the tip of the bones.
[SSCE 2005 Qu 50]

19. The structures which hold bones together at the joints are called
A. cartilages.
B. intervertebral disc.
C. ligament.
D. tendons.
[WASSCE 2006 Qu 11]

20. Which of the following happens when a human lifts a cup of tea from a table?
A. The triceps muscles contracts and the biceps relaxes
B. The biceps muscles relaxes and the triceps contracts.
C. Both biceps and triceps muscles contract.
D. The biceps muscle contract and the triceps relax.
[WASSCE 2006 Qu 12]

Use the following features of vertebrae to answer Questions **21** *and* **22**.

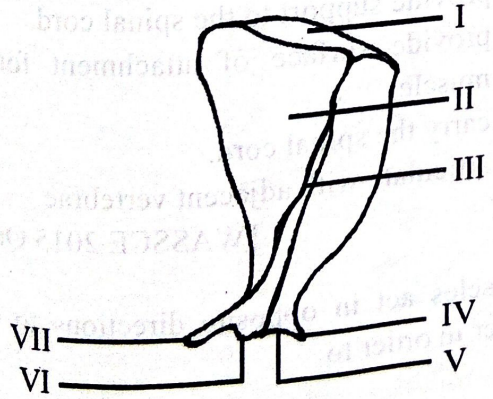
I. Centrum absent
II. Wing-like transverse process
III. Very long neural spine
IV. Odontoid process present
V. Very short transverse process

21. Which of the features listed above are characteristics of the atlas vertebra?
A. I and II only B. I and IV only
C. II and V only D. III and IV only
[WASSCE 2006 Qu 13]

22. Which of the listed features can be used to identify a thoracic vertebra?
A. I and V only B. I and IV only
C. III and V only D. IV and V only
[WASSCE 2006 Qu 14]

23. In humans, bending of the arm is brought about by the
A. contraction of both the biceps and triceps muscles.
B. contraction of the biceps and relaxation of triceps muscles.
C. relaxation of the biceps and relaxation of the triceps muscles.
D. relaxation of both the biceps and triceps muscles.
[WASSCE 2008 Qu 5]

Study the diagram below and use it to answer Questions **24** *to* **27**.



24. The structure represented in the diagram is called
A. clavicle. B. scapula.
C. sacrum. D. humerus.
[WASSCE 2008 Qu 6]

25. The part labelled **I** is the
A. supra-scapula. B. blade of scapula.
C. spine of scapula. D. glenoid cavity.
[WASSCE 2008 Qu 7]

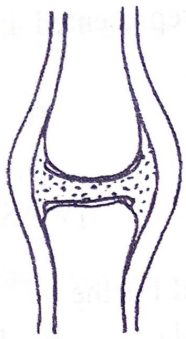
26. The part labelled **V**
A. articulates with the head of the humerus.
B. articulates with the olecranon process.
C. protects blood vessels and nerves.
D. provides large surface area for the attachment of muscles.
[WASSCE 2008 Qu 8]

27. The part labelled **VII** is the
A. coracoid process.
B. acromion process.
C. glenoid cavity.
D. metacromion process.
[WASSCE 2008 Qu 9]

28. The vertebrarterial canal is characteristic of the
A. thoracic vertebra. B. lumbar vertebra.
C. cervical vertebra. D. caudal vertebra.
[WASSCE 2008 Qu 10]

29. The following bones make up the axial skeleton **except** the
A. skull. B. vertebral column.
C. sternum. D. limb girdles.
[WASSCE 2009 Qu 10]

30. Which of the following tissues limits the extent to which bones can move at a joint?
A. Cartilage B. Ligaments
C. Tendons D. Muscles
[WASSCE 2012 Qu 8]



31. The diagram above is an illustration of a typical joint in which the synovial capsule and fluid have been removed. What will be the **greatest** effect on the joint?

A. Degeneration
B. Pain
C. Immobility
D. Retarded growth

[WASSCE 2012 Qu 9]

32. The two innominate bones are fused together ventrally at the

A. obturator foramen.
B. sacro-iliac joint.
C. pubic symphysis.
D. coracoid process.

[WASSCE 2013 Qu 7]

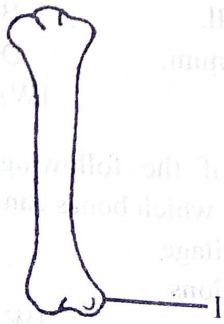
33. Two bones in the mammalian body are joined by means of

A. cartilage.
B. ligaments.
C. tendons.
D. muscle.

[WASSCE 2013 Qu 59]

The diagram below is an illustration of a mammalian bone.

Use it to answer Questions **34** *and* **35**.



34. The bone illustrated above is the

A. femur.
B. humerus.
C. radius.
D. ulna.

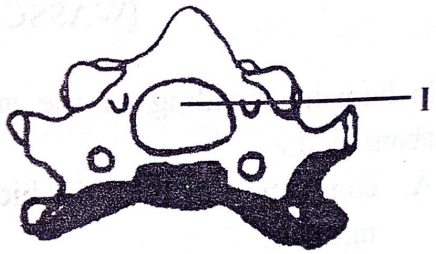
[WASSCE 2014 Qu 10]

35. The function of the part labelled **I** is to

A. fit into the glenoid cavity of the scapula.
B. fit into the acetabulum of the pelvic girdle.
C. support the lower arm.
D. prevent the arm from bending backwards.

[WASSCE 2014 Qu 11]

Use the diagram below illustrating a part of the mammalian skeleton to answer Questions **36** *and* **37**.



36. The part of the mammalian skeleton illustrated above is the

A. atlas vertebrae.
B. axis vertebrae.
C. cervical vertebrae.
D. thoracic vertebrae.

[WASSCE 2015 Qu 13]

37. The function of the part labelled **I** is to

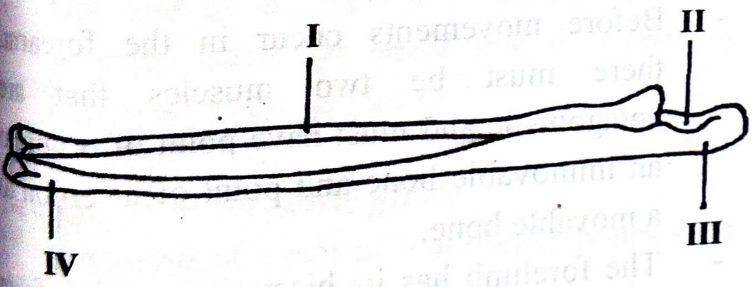
A. provide support to the spinal cord.
B. provide surface of attachment for the muscle.
C. carry the spinal cord.
D. articulate with adjacent vertebrae.

[WASSCE 2015 Qu 14]

38. Muscles act in opposite directions to each other in order to

A. cause a bone to move.
B. prevent dislocation at joints.
C. prevent muscle fatigue.
D. regulate bodily activities.
[WASSCE June 2016 Qu 11]

The diagram below illustrates a part of the human skeleton.
Study it and answer questions **39** *to* **41**.



39. The diagram represents the bones of the
A. upper arm. B. lower arm.
C. upper leg. D. lower leg.
[WASSCE June 2017 Qu 8]

40. Which of the labelled parts articulates with the head of the trochlea to form a hinge joint?
A. I B. II C. III D. IV
[WASSCE June 2017 Qu 9]

41. The labelled part that provides surface for the attachment of the triceps is
A. I B. II C. III D. IV
[WASSCE June 2017 Qu 10]

42. One of the structural adaptations of mammals for movement on land is the development of the
A. hollow bones.
B. pentadactyl limb.
C. fused thoracic bones.
D. streamlined body.
[WASSCE June 2017 Qu 48]

43. The odontoid process is found on the
A. sacral vertebra. B. atlas vertebra.
C. thoracic vertebra. D. axis vertebra.
[WASSCE June 2018 Qu 9]

44. The main function of the compressible cartilage pads in bones is to
A. prevent movement. B. produce fluids.
C. provide nutrients. D. prevent friction.
[WASSCE Nov 2019 Qu 10]